TW LAB

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Assignment 3

Que 1

$$\overbrace{a+b+c}^{FirstGroup}+\underbrace{x+y+z}_{SecondGroup}=\overbrace{\underbrace{a+b}_{Sum1}+c}^{Total}$$

Que 2

$$A=\left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Que 3

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x+1}, & 0\le x<4, \\ \dfrac{x^2-1}{x-1}, & x\ge 4. \end{cases}$$

Que 4

$$\left[\begin{array}{ccccc} \overbrace{a_{11} & a_{12} & a_{13}}^{\text{Column Group 1}} & \overbrace{a_{14} & a_{15}}^{\text{Column Group 2}} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{array}\right]$$

Que 5

$$F(x)=\underbrace{\overbrace{\sum_{i=1}^{n}(a_ix^2+b_ix_i+c_i)}^{QuadraticContribution}+\overbrace{\begin{bmatrix}x_1 & x_2 & \cdots & x_n\end{bmatrix}A\begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix}}^{MatrixContribution}}_{TotalEnergy}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

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Roll No.	Name	Department	Semester	CGPA
101	*Amit*	*CSE*	2	8.45
102	*Neha*	*CSE*	2	9.12
103	*Rahul*	*ECE*	4	8.76
104	*Priya*	*ECE*	4	9.01

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Academic Performance		
Student	**Midterm**	**Final**
A	78	89
B	85	92
C	91	94

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Department	Course	Credits	Grade
CSE	*DataStructures*	4	*A*
	MachineLearning	4	*A+*
	OperatingSystems	3	*B+*
ECE	*DigitalElectronics*	4	*A*
	Signals	3	*B+*

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Model	Parameters	FLOPs	Accuracy	Energy (kWh)
$ResNet18$	$11.7M$	$1.8G$	94.2%	0.00093
$MobileNetV2$	$3.5M$	$0.3G$	92.8%	0.00061
$EfficientNet-B0$	$5.3M$	$0.39G$	93.5%	0.00072

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Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

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Technique	Description	Advantages	Limitations
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

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Table 1: Scientific Model Comparison

Model	Parameters	FLOPs	Accuracy	Energy
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

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Model	Architecture	Performance	
CNN	Conv → ReLU → Pool	Metric	Value
		Accuracy	94%
		F1 Score	93%
ResNet	Conv → BN → ReLU	Metric	Value
		Accuracy	96%
		F1 Score	95%
ViT	Attention → MLP	Metric	Value
		Accuracy	97%
		F1 Score	96%

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Model	Loss Function	Parameters	Complexity
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	$10K$	$O(n)$
CNN	$L = -\sum y \log(\hat{y})$	$1.2M$	$O(n^2)$
Transformer	$L = -\sum_i y_i \log p_i$	$110M$	$O(n^2 d)$